

## LAB 6 REPORT

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**Lab Title:** Sets & Dictionary

**Theory:** **Sets** in Python are unordered collections of unique elements. They are used to store multiple values without duplicates and support operations like union and intersection. **Dictionaries** store data in **key-value pairs**. They are used to access and manage data using unique keys instead of indexes.

### TASK 1:

```
numbers = [1, 2, 2, 3, 4, 4, 5]
unique = set(numbers)
print(unique)
```

Output:

```
{1, 2, 3, 4, 5}
Process finished with exit code 0
```

### TASK 2:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

```
print("Union:", A | B)
print("Intersection:", A & B)
```

Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Process finished with exit code 0
```

### TASK 3:

```
student = {  
    "name": "Ali",  
    "age": 20,  
    "department": "CYS"  
}
```

```
print(student["name"])  
print(student["age"])  
print(student["department"])
```

Output:

```
↑  
↓  
↩  
↪  
⏎  
Process finished with exit code 0
```

#### TASK 4:

```
student = {  
    "name": "Ahmed",  
    "marks": 85  
}
```

```
student["marks"] = 90 # update  
student["city"] = "Lahore" # add  
del student["name"] # delete
```

```
print(student)
```

Output:

```
{'marks': 90, 'city': 'Lahore'}  
  
Process finished with exit code 0
```

